



Space Debris Classification - ML Project Documentation

Project Title: Space Debris Classification using Machine Learning **Objective:** Classify space objects into categories (DEBRIS, PAYLOAD, ROCKET BODY, TBA) using orbital parameters **Best Model:** XGBoost Classifier with 91.05% accuracy **Dataset:** 14,372 space objects with 40 orbital features

Project Structure Overview

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idse project/
├── Data Files
├── Trained Models
├── Analysis Notebooks
├── Core Pipeline Code
└── Configuration Files
```

Data Files Section

Purpose: Store and manage all datasets used in the ML pipeline

File	Purpose	Description
space_decay.csv	Original Dataset	Raw space object data (14,372 records × 40 features) containing orbital parameters like eccentricity, inclination, mean motion, etc.
artifacts/train.csv	Training Data	80% of data used for model training (11,497 records)
artifacts/test.csv	Test Data	20% of data reserved for final model evaluation (2,875 records)
notebook/data/space_decay_cleaned.csv	Cleaned Dataset	Preprocessed data after EDA with missing values handled and outliers removed

Goal: Provide clean, organized data for reproducible machine learning experiments



Trained Models Section

Purpose: Store all trained models and preprocessing objects for deployment

File	Purpose	Description
<code>notebook/data/best_model.pkl</code>	Production Model	Trained XGBoost classifier achieving 91.05% test accuracy
<code>notebook/data/scaler.pkl</code>	Feature Scaler	StandardScaler object for normalizing numerical features
<code>notebook/data/label_encoders.pkl</code>	Label Encoders	Encoders for categorical variables (COUNTRY_CODE, CLASSIFICATION_TYPE)
<code>artifacts/preprocessor.pkl</code>	Data Preprocessor	Complete preprocessing pipeline for new data

Goal: Enable immediate deployment and prediction on new space object data



Analysis Notebooks Section

Purpose: Document the complete data science workflow with interactive analysis

File	Purpose	Key Contents
<code>notebook/1_EDA_Analysis.ipynb</code>	Exploratory Data Analysis	Data quality assessment, Statistical summaries, Visualization of orbital patterns, Class distribution analysis, Missing value treatment, Outlier detection
<code>notebook/2_Model_Training.ipynb</code>	Model Development	Feature engineering, Multiple algorithm comparison, Hyperparameter tuning, Cross-validation, Performance evaluation, Model selection rationale

Goal: Provide transparent, reproducible analysis that can be presented to stakeholders



Core Pipeline Code Section

Purpose: Production-ready, modular code for the entire ML pipeline

src/components/ - Data Processing Components

File	Class/Function	Purpose
data_ingestion.py	DataIngestion	Load raw space object data, Perform train-test split, Save processed datasets, Handle data validation
data_transformation.py	DataTransformation	Feature scaling and normalization, Categorical encoding, Missing value imputation, Create preprocessing pipelines
model_trainer.py	ModelTrainer	Train multiple ML algorithms, Hyperparameter optimization, Model evaluation and comparison, Save best performing model

src/pipeline/ - End-to-End Workflows

File	Class/Function	Purpose
train_pipeline.py	TrainPipeline	Orchestrate complete training workflow, Connect all components sequentially, Handle error management, Log training progress
predict_pipeline.py	PredictPipeline	Load trained models for inference, Process new space object data, Generate predictions with confidence, Handle real-time classification

src/ - Utility Modules

File	Purpose	Key Functions
exception.py	Error Handling	Custom exception classes for debugging and error tracking
logger.py	Logging System	Structured logging for monitoring pipeline execution
utils.py	Helper Functions	Model saving/loading, data validation, common utilities
feature_engineering.py	Feature Creation	Advanced feature engineering for orbital mechanics

Goal: Create maintainable, scalable code that follows software engineering best practices



Configuration Files Section

Purpose: Project setup, dependencies, and execution scripts

File	Purpose	Contents
requirements.txt	Dependencies	All Python packages needed (pandas, scikit-learn, xgboost, etc.)
setup.py	Package Setup	Project metadata and installation configuration
run_pipeline.py	Main Executor	Entry point to run the complete ML pipeline
README.md	Project Guide	Installation instructions, usage examples, project overview

Goal: Enable easy project setup and execution for any user



Machine Learning Workflow

1. Data Pipeline

Raw Data → Data Ingestion → Train/Test Split → Feature Engineering → Cleaned Data

2. Model Development

Cleaned Data → Multiple Algorithms → Hyperparameter Tuning → Cross-Validation → Best Model



3. Production Deployment

Best Model → Model Serialization → Prediction Pipeline → Real-time Classification



Key Technical Achievements

Metric	Value	Significance
Model Accuracy	91.05%	Excellent performance for 4-class classification
Cross-Validation	91.11% ± 0.54%	Consistent, reliable performance
F1-Score	91.45%	Balanced precision and recall
Training Time	< 5 minutes	Efficient for production retraining
Prediction Speed	< 1 second	Real-time classification capability



Scientific Methodology

Problem Formulation

- **Input:** Orbital parameters (eccentricity, inclination, mean motion, etc.)
- **Output:** Space object classification (DEBRIS, PAYLOAD, ROCKET BODY, TBA)
- **Approach:** Supervised multi-class classification

Data Science Process

1. **Data Collection:** Space-Track.org API data 2. **Exploratory Analysis:** Statistical analysis and visualization 3. **Feature Engineering:** Orbital mechanics-based features 4. **Model Selection:** Comparison of 5 algorithms 5. **Validation:** Rigorous cross-validation and testing 6. **Deployment:** Production-ready pipeline

Technical Innovation

- **Class Imbalance Handling:** Balanced class weights for minority classes
- **Feature Engineering:** Domain-specific orbital mechanics features
- **Pipeline Architecture:** Modular, reusable components
- **Model Interpretability:** Feature importance analysis



Business Value & Applications

Space Industry Applications

- **Space Traffic Management:** Automated debris tracking
- **Satellite Operations:** Collision avoidance planning

- **Mission Planning:** Risk assessment for new launches
- **Insurance:** Space asset risk evaluation

Technical Benefits

- **Automation:** Reduces manual classification effort
 - **Accuracy:** 91% accuracy vs. human classification
 - **Speed:** Real-time processing of new objects
 - **Scalability:** Handles thousands of objects efficiently
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Educational Value

Demonstrates Mastery Of:

- **Data Science Workflow:** Complete end-to-end project
- **Machine Learning:** Multiple algorithms and evaluation
- **Software Engineering:** Clean, modular code architecture
- **Domain Knowledge:** Space science and orbital mechanics
- **Statistical Analysis:** Rigorous validation methodology

Skills Showcased:

- Python programming and ML libraries
 - Data visualization and analysis
 - Model selection and hyperparameter tuning
 - Production deployment considerations
 - Technical documentation and presentation
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 Project Statistics

Component	Count	Details
Data Points	14,372	Space objects analyzed
Features	15	Orbital parameters used
Models Trained	5	Algorithms compared
Code Files	12	Production-ready modules
Notebooks	2	Interactive analysis
Accuracy	91.05%	Final model performance

 Conclusion

This project demonstrates a **complete, professional-grade machine learning solution** for space debris classification. It combines:

- **Strong Technical Skills:** Advanced ML techniques and software engineering
- **Domain Expertise:** Understanding of orbital mechanics and space science
- **Practical Impact:** Real-world application with measurable business value
- **Academic Rigor:** Proper methodology and thorough validation

The project is **production-ready, well-documented**, and **scientifically sound**, making it an excellent demonstration of data science capabilities for academic and professional evaluation.

Ready for Presentation

This documentation is ready for presentation to professors, industry professionals, or technical stakeholders! 🎯